RADIOS (CAT) — CI-V

IC-820 0x42/9600 · **IC-821** 0x4C/9600 · **IC-910** 0x60/19200 · **IC-970** 0x2E/9600 · **IC-9100** 0x7C/19200 · **IC-9700** 0xA2/19200. Address & baud per-rig configurable in Settings. Single-wire CI-V confirmed on IC-821.

RADIOS — Yaesu / Kenwood CAT

FT-847 57600 · **FT-736R** 4800 · **TS-790** 4800 · **TS-2000** 57600.

CAT TRANSPORTS (pick in Settings)

Wired TTL UART CI-V (single or separate pin) · **Icom LAN** RS-BA1 UDP 50001/50002 (IC-9700) · **rigctl (net)** Hamlib NET TCP (VFOA=DL/B=UL) · **rigctl (Grove)** same protocol over Grove G1/G2, no Wi-Fi (port row = baud) · **USB** adapter on USB-C: FTDI 0403, CP210x 10c4, CH34x 1a86, PL2303 067b, any CDC-ACM. USB device strings lead with **#N** = device address = the id explicit binding stores (tells identical adapters apart).

DUAL-RIG + TRANSVERTER

CardSatDualRig (M5StickS3 companion): two half-duplex/RX radios as one full-duplex station over a rigctld server; drive via rigctl (net/Grove). **Settings→Radio→Dual-Rig setup** shows the Stick's live USB enumeration to bind a device per leg (link dot red/green). **d** gains **FULLu** = One True Rule on the UPLINK knob. **Transverter LO**: Downlink/Uplink LO (MHz) drive a microwave up/down-converter (rig on IF = real–LO; DN/UP show **x**). Both paths **UNTESTED on hardware** — verify.

ROTATORS

Serial protocols: **GS-232A/B**, **Easycomm I/II/III**, **SPID Rot2Prog** — each over the **I2C bridge** (SC16IS750), **Grove G1/G2**, or a **USB adapter** (Rot wire setting). Network: **rotctld** TCP, **PstRotator** UDP. Direct **Yaesu** via I2C ADC + expander (needs calibration). Grove rotator excludes wired CAT and Grove GPS (one UART).

DATA SOURCES

GP elements (JSON): **AMSAT** nasabare (default), any **CelesTrak** group, or a custom URL; plus whole-catalog CelesTrak **search** by NAME or CATNR. **SatNOGS** transponders · **AMSAT status** API · **hams.at** activations · **LoTW** upload · Cloudlog · **Open-Meteo** weather (CC BY 4.0) · **NOAA SWPC** space weather. Satellite names resolve source-independently (parenthetical designator bridge).

COURTESY LIMITS (automatic)

Primary GP catalog & CelesTrak extras re-fetch no sooner than **2 h** per source URL (persisted across reboots; changing source fetches at once) · catalog search **10 s** between queries · extras capped at **25**.

CAPACITIES

150 satellites resident (favorites survive truncation) · **64** transponders for the active sat · **25** CelesTrak extras · BASIC: **500,000** statements/run, **2,000** SATSEL calls, @() array ≤ **256**, **6 KB** output — and **no INPUT**, by design.

ORBIT ENGINE

SGP4/SDP4 (deep space included). Periods over ~225 min (Molniya, GTO, GEO) use the scan finder: ~1 s AOS/LOS, horizon-long pass for a bird parked in view — Skyfield-verified (crossings ≤ 0.04°). Doppler: AMSAT One True Rule, per-satellite calibration.

KEY FILES (/CardSat/ on SD)

config.json(+.bak) settings · **gp.json** primary catalog · **mgp.json** hand-entered sats · **ctq.json/ctx.json** search results / extras (+**ctx.ts, gp.ts** courtesy timestamps) · **basic/** programs & gated logs · **plot.csv** grapher CSV mode · **Reports/** printed reports · **Logs/** QSO log · **RovePlans/** · **Screenshots/** · **calib.txt** per-sat cal · **lotw_sats.csv** LoTW name overrides · **audio/** voice memos · **telnet.txt** saved Telnet connections (label|host|port|prnCols|outMode).

CALCULATOR — conventions

Trig in **degrees** · **Ans** = last result · **x** = variable in the grapher · suffixes on numbers: **f p n u m k M G T** (case matters: M mega, m milli) · \ toggles engineering-notation output.

General functions

sin cos tan asin acos atan · atan2(y,x) · sinh cosh tanh · ln log log2 exp · sqrt cbrt · abs floor ceil round sign · fact(n) ncr(n,r) npr(n,r) · min max mod hypot (2-arg) · rnd() · d2r r2d.

Ham / RF functions

dbm(W) w(dBm) · db(ratio) undb(dB) · wl(MHz)→m fq(m)→MHz · swr2rl rl2swr · mml(swr) mismatch loss · fspl(MHz,km) path loss · nf2t(dB) t2nf(K) noise fig ↔ temp · dbd dbi (±2.15) · dop(MHz, rrKmS)→Hz — feed it SATRR.

Orbital one-liners + constants

porb(altKm)→min · vorb(altKm)→km/s · fpr(altKm)→footprint km. Constants: pi e c kB Re mu g0.

TINY BASIC — statements

LET / bare **V=** · PRINT (**?**) · IF..THEN · GOTO · GOSUB/RETURN · FOR..TO..[STEP]..NEXT · ON e GOTO n1,n2,.. · DIM @(n) · DATA / READ / RESTORE · REM · END · **SATSEL i** re-snapshot SAT* for catalog sat i · **TXSEL i** → TX* · **LPRINT** to report sinks · gfx: **CLS PSET LINE CIRCLE TEXT SHOW** (colors 0-9; SHOWed frame holds after the run) · **FOPEN"n" FPRINT FCLOSE** (Settings-gated, /CardSat/basic/) · FILES.

Expressions & functions

Ops: + - * / ^ **MOD**; IF compares = <> < > <= >= with **AND OR NOT**. Vars A-Z + one **@(i)** array. Fns: ABS INT SQR SGN SIN COS TAN ATN LOG EXP MIN MAX RND(n). Trig in degrees, matching the calculators.

System names (read-only snapshot)

Sat: SATAZ EL RNG RR LAT LON ALT SUN INC ECC RAAN MM NOR · **Pass**: AOSIN LOSIN PASSEL PASSVIS PASSN PASSAOS/LOS/MAX(k≤8) · **Sky**: SUNAZ SUNEL MOONAZ MOONEL · **Site**: MYLAT MYLON MYALT LSTHR · **Clock**: UTCYR MON DAY H M S · **SpaceWx**: SFI KP AINDEX · **Wx**: WXTEMP WIND DIR HUM · **GPS**: GPSSATS GPSSPD GPSLAT LON ALT · **Device**: BATT GPAGE NFAV HEAPFREE UPTIME NSAT NTX · **TX** (after TXSEL): TXDL UL BW INV LIN · **Flags**: SATOK TIMEOK POSOK WXOK SPWXOK PASSOK GPSOK.